EMANUEL MARROQUÍN

FULL STACK DEVELOPER — .NET, PYTHON, APPLIED AI

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Check out my portfolio

SUMMARY

Full stack developer with 4+ years of experience building and modernizing web applications end to end: backend in .NET, Node.js and Python; frontend in React, Next.js and TypeScript. Currently migrating legacy products to .NET 8 in a high-availability enterprise financial environment, deployed on Kubernetes and OpenShift. Pursuing a Master's in Information Security, with a focus on security by design (OWASP), sound architecture and observability. Builds and ships applied AI systems (RAG, LLM serving, agent tooling).

EXPERIENCE

Full Stack Developer

Canella, S.A. Apr. 2025 – Present

- Migrate legacy products (C# / ASP.NET) to .NET 8 in a high-availability enterprise financial environment, modernizing business-critical code.
- Deploy and maintain containerized services with Docker, Kubernetes and OpenShift.
- Work with CI/CD pipelines in Azure DevOps and GitHub.
- Use AI coding assistants daily to speed up the migration, reviewing all generated code critically before it ships.

Freelance Full Stack Engineer

Independent Jan. 2023 – Mar. 2025

- Designed and built web systems for clients (land-lot management, agricultural order management).
- Built responsive React frontends (reusable components, state management) and Node.js / Express / MongoDB backends.
- Owned the full lifecycle: requirements gathering, development, deployment and maintenance.

SELECTED PROJECTS

Aegis-RAG Python, FastAPI, Docker, AWS, Terraform — Aegis-RAG Github

- Security-first RAG API with hexagonal architecture: every query passes through a security gateway before retrieval or generation, with mitigations for the OWASP LLM Top 10.
- Full observability (logs, Prometheus metrics, OpenTelemetry traces) and unit, integration and end-to-end tests (~85% coverage).

CodeSentinel C#, .NET 8, CI/CD — CodeSentinel Github

- Repository security scanner that detects exposed secrets and insecure patterns, with pluggable rules and a score for CI/CD gating. JSON, HTML and SARIF reports.

Nexus AI Orchestrator TypeScript, React, Next.js, Tailwind CSS — Nexus AI Github

- Visual node-based editor for AI pipelines: Zustand slice-based state, live telemetry over server-sent events (SSE), undo/redo, and unit tests with Vitest.

Quetzal-lakehouse Python, dbt, DuckDB, Airflow, Terraform — Quetzal Github

- Lakehouse (bronze/silver/gold) on Guatemalan public financial data: idempotent ingestion from three sources, dbt star schema with data tests, daily Airflow orchestration and Terraform-defined AWS storage.

SKILLS

- **Backend:** C#, .NET 8, ASP.NET Core, Entity Framework Core, Node.js, Express, Python, FastAPI, REST APIs
- **Frontend:** React, Next.js, TypeScript, JavaScript, Tailwind CSS, HTML5, CSS3
- **Databases:** SQL Server, PostgreSQL, MongoDB, Redis
- **Data:** dbt, DuckDB, Airflow, dimensional modeling, Parquet
- **Architecture:** Hexagonal architecture, microservices, dependency injection, technical documentation (ADRs)
- **Cloud & DevOps:** Kubernetes, OpenShift, Docker, Azure DevOps, CI/CD, AWS, Terraform, Prometheus, OpenTelemetry
- **Security:** OWASP Top 10, OWASP LLM Top 10, security by design, ISO 27001:2022, COBIT 2023
- **Applied AI:** AI-assisted development (Claude Code), review of AI-generated code, prompt engineering, RAG, agents (LangGraph), LLM guardrails

LANGUAGES

- English (B2+)
- Spanish (native)

EDUCATION

Bachelor's in Systems Engineering

Universidad Mariano Gálvez (2020-2024)
coursework completed; degree pending

Master's in Information Security

Universidad Mariano Gálvez (2025-Present.)
final term

Claude Code in Action (Anthropic)

Scrum Fundamentals Certified (SCRUM)

AWS Cloud Practitioner Essentials (AWS)

JavaScript Algorithms and Data Structures